Arista Access Node WFD-ACC-14 - Optical Transport Failure Resolution SOP

Node: WFD-ACC-14 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_5573

Description

This SOP outlines the procedures for resolving an Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Arista Access Node WFD-ACC-14 at the Whitefield POP. On Arista platforms, access nodes are part of the network fabric, utilizing the Transparent Interconnection of Lots of Links (TRILL) protocol for multipathing and load balancing. In the event of an Optical Transport Failure, the failure domain is triggered due to the degradation or loss of the received or transmitted optical signal, impacting the node's ability to forward traffic. The root cause of such failures can be attributed to issues with the SFP/XFP modules, fiber cuts, or power loss in the optical transport path. This SOP aims to provide a structured approach to diagnosing and resolving such failures, ensuring minimal downtime and maintaining the integrity of the network.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Whitefield POP for physical access and assistance.

- Verify the status of redundant links to determine if the failure is isolated to a single link or a more extensive network issue.

- Confirm the node's configuration and ensure that there are no software or firmware issues contributing to the failure.

Potential Root Causes

Cause	Probability
SFP/XFP Module Degradation	High
Fiber Cut	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware sfp, Expected Output: SFP information (e.g., SFP type, status, and temperature), Description: This command displays information about the SFP/XFP modules, including their type, status, and temperature. It helps identify potential issues with the modules or the optical transport path.

Command: show interface tengigabitethernet 1/1/1, Expected Output: Interface information (e.g., speed, duplex, and status), Description: This command provides information about the interface, including its speed, duplex, and status. It helps determine if the interface is functioning correctly and if there are any issues with the optical transport path.
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```
Command: show log | include ALARM_5573, Expected Output: Alarm log entries (e.g., timestamp, severity, and description), Description: This command displays alarm log entries related to the Optical Transport Failure, including the timestamp, severity, and description. It helps identify when the failure occurred and its impact on the network.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP Module, Command: replace hardware sfp slot 1/1/1, Description: Replace the SFP/XFP module in slot 1/1/1 with a new one. Ensure the new module is properly seated and secured.

Step 2: Step: Verify Interface Status, Command: show interface tengigabitethernet 1/1/1, Description: Verify the status of the interface after replacing the SFP/XFP module. Ensure the interface is functioning correctly and there are no issues with the optical transport path.

Step 3: Step: Clear Alarms, Command: clear log alarm, Description: Clear the alarm log entries related to the Optical Transport Failure. This will prevent the alarm from reappearing and ensure the network returns to a stable state.

Post-Check Verification

```
Command: show hardware sfp, Expected Output: SFP information (e.g., SFP type, status, and temperature), Description: Verify that the SFP/XFP module has been replaced successfully and is functioning correctly.

Command: show interface tengigabitethernet 1/1/1, Expected Output: Interface information (e.g., speed, duplex, and status), Description: Verify that the interface is functioning correctly and there are no issues with the optical transport path.

Command: show log | include ALARM_5573, Expected Output: No alarm log entries related to the Optical Transport Failure, Description: Verify that the alarm log entries related to the Optical Transport Failure have been cleared.
```

Roll-back Procedures

- Step: Restore Original SFP/XFP Module, Command: replace hardware sfp slot 1/1/1 original, Description: Restore the original SFP/XFP module in slot 1/1/1. Ensure the module is properly seated and secured.

- Step: Re-enable Alarms, Command: enable log alarm, Description: Re-enable the alarm log entries related to the Optical Transport Failure. This will ensure the network returns to its previous state.